What happens if two files in the same directory have similar names, like Caina1.doc and Caina2.doc? In that case, tab completion will sort through the files in alphabetical order. Say you typed this command and then hit the TAB key:

DEL Caina

The Command Prompt would first produce this:

DEL Caina1.doc

But if you hit the TAB key for a second time, it would switch to this:

DEL Caina2.doc

With tab completion, you can type long filenames far quicker than you could otherwise.

FILENAMES WITH SPACES

In the DOS era, you couldn't have filenames or directory names with spaces. But as we've mentioned, in modern versions of Windows, you can have filenames up to 255 characters long, including spaces. In fact, you can have any characters you want in a filename, except for a forward slash (/), a backslash (\), a colon (:), an asterisk (*), a less than sign (<), a greater than sign (>), and a pipe (|). The

reason is that many of these characters serve as filename wildcards and command redirection characters, which we shall discuss in greater detail in Chapter 6.

But back to filenames with spaces. The Mac OS X and the Linux command line require you to denote spaces in file names with slashes or quotation marks, since otherwise their command-line interpreters will not parse the spaces properly. The Windows Command Prompt, however, can handle spaces just fine. For instance, if you want to change to the Program Files directory, you would simply type this command:

CD \Program Files

And with tab completion, you can quickly fill out "Program Files" without having to type the entire name.

There are certain circumstances, such as batch files, where using the full name with spaces is undesirable. For every long file name, Windows also generates a hidden 8.3 filename, since certain applications and situations require a shorter file name. We'll discuss how to find those short 8.3 filenames when we discuss the DIR command in detail in Chapter 4.

COPYING AND PASTING

Many Command Prompt commands are long and complicated. Even with tab completion, typing them out can be a chore. Copying and pasting complex commands into the Command Prompt would make things easier - both the Mac OS X and Linux terminal applications include Edit menus in their menu bars that

allow users to paste commands into the prompt and copy the output of the commands. Why not do the same with Windows?

Unfortunately, this is one area where the Windows Command Prompt falls behind the command-line applications of competing operating system. The Command Prompt does not have the typical Windows menu bar, which means it has neither the Edit menu nor the Copy and Paste items. To make matters worse, the standard CTRL+C shortcut for Copy and the CTRL+V shortcut for Paste do not work in Command Prompt.

That said, it is possible to copy text from and paste text into the Command Prompt. It just takes a little more work than most other Windows applications. We'll first discuss how to paste items into the Command Prompt, and then how to copy text from the prompt.

The Command Prompt doesn't have the usual Windows menu bar, but it does have a menu of its own in the upper-left hand corner of the title bar, indicated by the small Command Prompt icon. Left-click on this icon to summon the Command Prompt menu, and you'll see an option for Edit about two-thirds of the way down. Move your mouse pointer over that, and the menu will expand to show the usual Edit options, including Paste, along with a few others. Click on Paste, and any text currently on your clipboard (such as a copied command) will be pasted into Command Prompt.

Copying from the Command Prompt is a bit trickier. By default, you can’t highlight text within the Command Prompt window. To highlight text, go to the Command Prompt menu, to the Edit category, and then click Mark. You’ll now be able to highlight text within the Command Prompt windows. Next, go back to the menu, to Edit, and select Copy – the highlighted text will be copied to your clipboard. (While text is highlighted, you can also use the Enter key to copy.) You can then Paste it into the application of your choice – this is especially useful when you’re getting an error message, and you want to email it to a

technician or post it to a forum.

Copying text in this way is rather cumbersome. There is a way to speed it up. If you right-click on Command Prompt's title bar and click on Properties, this will take you to the Properties dialog box. Most of the Properties dialog box's tabs let you control the shape, color, and font size of the Command Prompt. But under the options tab, there is a check box labeled "Insert Mode." If you check that, it will be as if the Mark option is turned on in Command Prompt all the time – you can highlight text simply by left-clicking and dragging the mouse cursor over it, rather than going to the menu, selecting Edit, and then clicking on Mark every time you want to copy some text.

FINDING TEXT

During a long Command Prompt session, the window tends to fill up with a lot of text – command outputs, directory listings, and so forth. If you want to find a particular item, you can scroll back up through the session, but that can take a long time. The Command Prompt's menu includes a Find item, which lets you search the output of your commands for a particular word or phrase.

To use it, go to the Command Prompt menu in the upper-left hand corner, select the Edit item, and then click on Find. This will bring up a Find box, similar to the Find box in other Microsoft applications like Notepad and WordPad. Type the word or term you want to find into the search box and hit Find Next, and Command Prompt will find the term in the output of your session. Keep hitting Find Next to find additional instances of the term. You can also use the radio buttons under Direction to choose whether Command Prompt will search your term from the top of the screen down or from the bottom of the screen upward.

CLEARING THE SCREEN

During a busy Command Prompt session, the window will fill up with quite a bit of text. This can get rather cluttered, so you might want to clear the window entirely. Simply type this command at the prompt:

CLS

And your Command Prompt window will be cleared.

Note that all the text in the windows is gone forever after you use the CLS command, so if you want to copy and paste anything out the window, do so before using the cls command.

COMMAND HISTORY

As you use the Command Prompt, you might find yourself reusing the same commands, or similar commands, over and over again. You could of course retype them, but Command Prompt offers a feature called command history that makes retyping the commands unnecessary. Command history remembers the last fifty commands you type, and you can recall them to the prompt by pressing the UP arrow key on your keyboard. Keep pressing UP to scroll through the list of commands until you find the one you need. If you go too far into the list, you can use the DOWN arrow key to scroll backwards through the commands.

If command history doesn’t remember enough commands for your liking, you can adjust the number by right-clicking on Command Prompt’s title bar and selecting Properties. Under the Options tab, there is a heading for Command

History, and you can increase and decrease the buffer size for stored commands there.

CASE SENSITIVTY

If you've ever used the command line for Mac OS X or Linux, you know that their command line applications are case sensitive. That means their command interpreters distinguish between uppercase and lowercase letters. For instance, a Linux command line would view COPY, Copy, and copy as three separate commands. Likewise, a Linux command line would view Report.doc, REPORT.doc, and report.doc as three separate files.

Unlike Linux or Mac OS X, the Windows Command Prompt is not case sensitive, and does not distinguish between commands or filenames based up the case of the letters in the file name. To return to the previous example, the Command Prompt will interpret COPY, Copy, and copy as the same thing - every one of these will launch the copy command. In the same vein, Command Prompt will view Report.doc, REPORT.doc, and report.doc as the same file - it will not distinguish filenames based on the case of the letters in the filename.

This generally makes the Windows Command Prompt easier to use than the Mac OS X or Linux terminals, especially for new users.

For the sake of readability in this book, all commands will be typed in ALL CAPS, but it makes absolutely no difference whether you type the commands in uppercase, lowercase, or a mix of the two.

EXECUTABLE FILES

There are limitations to the kind of files you can open and programs you can launch from the Windows Command Prompt. You can open documents and image files from the Command Prompt - if you type the file name and hit enter, Windows will attempt to open the document using the default program for opening that type of file. If you type a filename and Command Prompt doesn't recognize it, you'll get this error message:

FILENAME is not recognized as an internal or external command, operable program or batch file.

(FILENAME, of course, is whatever you typed that Command Prompt doesn't recognize.)

So what does Command Prompt mean by "internal or external command, operable program, or batch file?" There are three different kinds of commands you can actually execute from the Command Prompt: COM files, EXE files, and BAT files. (There are actually more than three, but most people will only encounter these three kinds.)

A COM file is generally a "command" file, and refers to the commands included with Windows for use in Command Prompt. A large percentage of the commands we'll discuss in this book are actually COM files tucked away somewhere in your hard drive's \Windows directory. Occasionally commercial programs have a COM extension, but not very often.

Most commercial programs, and some Windows programs, are EXE files - executable files. Almost all programs written nowadays are stored in EXE files. Many of Windows's graphical applications have a file extension of EXE. You

can run some EXE files from the Command Prompt, but many EXE files can only be run in the graphical environment of Windows.

The final kind of executable file is a BAT file, a batch file. Batch files are different from COM and EXE files. Both COM and EXE files are binary files, and if you opened them with a text editor like Notepad, you would see only gibberish. Batch files, by contrast, are text files containing a list of commands. When you run a batch file from Command Prompt, it executes the commands in the batch file, one by one.

We'll discuss batch files more in Chapter 13.

GETTING HELP

Each command offers its own array of switches and options, and keeping track of them all can prove quite a challenge. Fortunately, every command supports the /? switch. Type any command with the /? switch appended, and Command Prompt will list the available options for the command. For example:

XCOPY /?

This will list all the options available for the XCOPY command.

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Chapter 2 - Drive Letters and the File System

If you've ever opened My Computer on Windows XP or Computer on Windows Vista and up, right away you've seen a large number of drives. Some of them might have been your computer's internal hard drives and optical drives. Others might have belonged to USB drives, like external hard drives or flash drives. Still others might have belonged to your computer's built-in media card reader, if your system has one. And if you have an old computer, you might even still have a letter assigned to a floppy drive, though floppy drives are becoming rarer and rarer. (Thankfully, in my opinion- floppy drives are slow and unreliable, and a $10 four gigabyte flash drive can hold the contents of roughly 1,500 three-and-a-half inch floppy disks.)

Before you can use the Command Prompt effectively, you need to understand how drive letters work, and how the Windows file system is laid out. We'll address both these topics in this chapter.

DRIVE LETTERS

In the Windows operating system, every physical drive (or logical partition on a physical drive – more on those in Chapter 9) is assigned a drive letter. For instance, your first hard drive might receive the letter C, your first optical drive the letter D, and so forth. This is different from Mac OS X and Linux, where additional drives generally show up as subfolders on the hard drive – usually in the /media or the /Volumes folder (a USB flash drive with a volume label of STORAGE would show up with a drive letter on a Windows system, but as the /Volumes/STORAGE directory on a Mac).

Why use drive letters? They provide a convenient way of finding which disk is holding a particular file. If you know that the full path to a file named Book.doc is C:\Storage\Book.doc file, then you know that the file is on the C drive, which is probably the first hard drive on your system.

Any drive can be assigned any letter. However, by tradition and convention (and in some cases technical requirements) certain letters usually get assigned to certain kinds of drives. The letters A and B are rarely used on modern Windows systems – in older DOS computers, the letters A and B were reserved for the first and second floppy drives in the system.

The letter C is almost always assigned to the first hard drive (or hard drive partition) in a system, generally the partition or hard drive where Windows is installed. On a modern computer, you can install Windows on any hard drive with any letter. However, it is usually best to install Windows on C, since many older (and badly-written) programs expect to find Windows on the C drive, and refuse to run if Windows is installed on any other drive letter.

The letter D, traditionally, is assigned to the first optical drive in the computer – a CD-ROM, DVD-ROM, CD-RW, or DVD-RW drive (or a drive that combines all four functions, as most modern drives do). However, on many modern laptops and desktops, the D drive is assigned to a recovery partition on the hard drive, which contains the files necessary to reinstall Windows from scratch in case of operating system corruption or irreparable virus damage. (Computer manufacturers started doing this to save on the cost of including a recovery DVD with the system.) If your system has a recovery partition on D, then your first optical drive will receive the letter of E.

After D and E, there is no generally agreed convention for which devices get which letters. As you add drives to your system, whether internal hard drives or removable USB flash drives and hard drives, Windows will assign them the next available letter. If you want a device to have a different letter, you can reassign

the letter either using the Disk Management snap-in in the Computer Management console, or using the DISKPART command-line utility (more on DISKPART in Chapter 9).

THE PATH

As you work with the Command Prompt, you might notice that in order to run an executable file you need to either change to the directory containing a file, or type its full path at the prompt. However, for many of the built-in Windows commands, you can type the command in any directory, and Command Prompt will run it. Why is that?

This is something called the system path, or generally just the path, that Command Prompt uses to find commands. Basically, the path consists of a few directories set apart by Windows for storing commands. If you type a command at the prompt, Command Prompt searches the directories in the path for the command. If it doesn't find the command in the path, it then checks the current directory. And if it doesn't find the command there, it comes back with the usual error message:

FILENAME is not recognized as an internal or external command, operable program or batch file.

If you want to find out what your system's path is, simply type the PATH command at the prompt:

PATH

The results will generally look something like this:

C:\Windows\system32;C:\Windows;C:\Windows\System32\Wbem

Command Prompt will search these three directories for commands. There might be other directories in your system's path – installed programs sometimes add their directories to the path. You can also add directories to the path using the PATH command, though it's best to do so sparingly. If you add too many commands to the path, Command Prompt will search them all to find a command, which can slow down system performance.

NETWORK DRIVES

From the days of Windows for Workgroups in the early 1990s, Windows has offered strong support for file sharing – accessing files stored in a shared folder on a remote server. It is possible to connect file shares to your computer and assign them a drive letter – a process called "drive mapping" or "mapping a network drive." Once a shared folder has been mapped to your computer as a network drive, you can use it like any other kind of drive, copying files to and from it (though that may depend on your permissions to the shared folder). We'll discuss network drives more in Chapter 10.

THE WINDOWS FILE SYSTEM

We've been talking about "directories", but what do we mean by that term?

If you look at the root of your C drive through Windows Explorer, you'll see a number of different folder icons. A "directory" is a virtual folder that allows you to sort and organize your files into different containers, so everything isn't messily stored in just one folder. (In fact, in earlier versions of Windows, if you had too many items in your hard drive's root directory, Windows mistakenly assumed that your hard drive had filled up!)

You could put additional directories inside directories – these directories within directories are called "subdirectories." You can put additional subdirectories within those subdirectories, as many as you wish. If you think of the root directory of your hard drive (represented by the \ character) as the trunk of a tree, then the directories and their subdirectories are branches off the main trunk. In fact, Windows includes a command that lets you view the filesystem as a tree:

TREE

Type that command at the prompt, and you will see your filesystem represented as a tree. It will quickly scroll past your screen – the modern Windows operating system contains many thousands of directories and subdirectories. In the next section, we'll talk about some of the more important directories on your computer's hard drive.

IMPORTANT DIRECTORIES

If you go to the root directory of your C drive (or the drive where Windows is installed), you should see a few directories. The most important of the directories is C:\Windows, which contains the system files that make up Windows itself. The Windows directory also contains the C:\Windows\system32 directory, which

holds a number of important components and utilities for Windows itself. Do not delete any of the files in this directory! Doing so can disable important parts of Windows, and may even render your system unbootable.

Also in the root directory of your C drive is the C:\Program Files folder. By default, your system's applications install themselves here. If your computer runs a 64-bit version of Windows, you'll also have a C:\Program Files (x86) folder, where Windows installs any 32-bit software programs. All 64-bit applications will go into the regular C:\Program Files folder on a 64-bit Windows system. If you need to uninstall a program, it's a bad idea to simply go into the C:\Program Files directory and start deleting files – you will get better results by uninstalling an application using the Programs and Features item in Control Panel.

The C:\Program Data directory is hidden, but contains a number of application data files and system data files. For instance, all the Start Menu shortcuts are stored in C:\ProgramData\Microsoft\Windows\Start Menu\Programs. Unless you know what you're doing, it's not a good idea to delete or alter files in the C:\Program Data directory.

The root directory of C also holds the C:\Users folder. This folder contains the user profiles – the personal data – of every user who logs into the system. Go into the C:\Users folder, and you will see folders named for every user account on the system. Inside each of those folders, you will see the profile folders for the user – their Desktop folder, which holds the files on their Desktop, their Documents folder, their Pictures folder, and so forth. There are also a number of hidden directories that contain application settings, data files, user preferences, and similar files.

Generally, when working on a computer, it is always a good idea to back up the contents of the C:\Users directory. Especially if it is the personal machine of a technically unskilled user – the user will almost always claim that he or she doesn't "have anything important" on the machine if you need to wipe the hard

drive and reinstall Windows. Nevertheless, if you do wipe the machine and reinstall Windows, a few weeks later the user will become irate when he notices that “all his stuff is missing” – and of course the user has no backup copies! It is therefore always a good idea to back up the C:\Users directory when repairing a Windows computer.

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Chapter 3 – Understanding the prompt

When you start the Command Prompt program, the first thing you'll see on your screen is this:

C:\>

That is the actual command prompt itself. But what does the prompt do? What do all those symbols mean? We'll discuss how to understand and alter the prompt in this chapter.

WHAT IS THE PROMPT?

The "prompt" that gives the Command Prompt its name is the first thing you see when you launch the Command Prompt program. What is it? To put it simply, its a series of characters that indicates the computer is ready to accept commands from you. Nowadays, you know that a computer has booted up and is ready to run programs when you see the Windows desktop. But in the days of DOS, you only knew the computer was ready when you saw the DOS prompt appear on the screen.

The basic prompt in the modern Command Prompt looks something like this:

C:\>

Alternatively, it can have additional directory paths within itself - generally, when you launch a new Command Prompt session, it opens in your home folder, so it will probably look something like this:

C:\Users\USERNAME>

What does this all mean?

If you remember the previous chapter on drive letters, the "C" in the prompt refers to the C drive, your computer's hard drive. You can change the selected drive in Command Prompt by typing the drive letter followed by a colon. For instance, if you wanted to switch from the C drive to the D drive, you would type this and his enter at the prompt:

D:

And then prompt would change from this:

C:\>

To this:

D:\>

If the first letter in the prompt indicates the currently selected drive, the characters after the drive show the current directory and subdirectory on the selected drive. In a prompt that looks like "C:\>", that indicates you are in the root directory of the C drive, since the "\" character represents the root directory of a drive. And as you change the current directory, the prompt changes as well to show the current directory. For instance, if you change from the root directory to the Program Files directory, the prompt will look like this:

C:\Program Files>

As you change into deeper and deeper subdirectories, the prompt will accordingly become longer and longer.

The final character in the prompt, the greater than sign (>), indicates the end of the prompt and the location of the cursor. Any commands you type into the prompt will appear to the right of the greater than sign.

CHANGING THE PROMPT

The default prompt configuration is good enough for most people. However, if you would prefer to customize the prompt to show additional (or less) information, you can do so with the PROMPT command. The PROMPT command lets you alter the appearance of the prompt, while customizing it however you wish.

To test it out, type this command at the prompt:

PROMPT Hello!

You'll notice that the command prompt immediately changes from "C:\>" to "Hello!" While having the prompt say "Hello!" seems quite friendly, it nonetheless contains no useful information. To change it back, type this command at the modified prompt:

PROMPT PG

The prompt will then change back to the more familiar "C:\>".

As you might guess, the PROMPT command lets you do more than simply put friendly (or snarky) messages in the command prompt. The PROMPT command responds to a series of variables, and will modify the prompt to display information based on those variables. In the above example, the $P variable refers to the current location, and the $G variable refers to the greater than sign. So when you issue a PROMPT command with the variables in this order, you'll return to the prompt to its default state:

PROMPT PG

Here are some the more useful variables you can use with the PROMPT command to enhance the default command prompt:

-$D adds the current date to the prompt.

-$C adds a left parenthesis.

-$F adds a right parenthesis. You can use this in conjunction with $D to place information provided by the other variables with a set of parentheses.

-$N displays the currently selected drive.

-$P displays both the currently selected drive and the currently selected location on that drive.

-$T shows the current time.

-$V shows the Windows version number.

You can get a complete list of variables for the PROMPT command by using it with the /? switch:

PROMPT /?

(As we mentioned earlier, using the /? switch with any command will provide a list of options for that command. It's a useful switch to remember, in case you can't recall the switches available for any one command.)